Ghazi Muhammad Abdullah

Domain: Artificial Intelligence & Machine Learning

Submitted to : Mr. Mohazin Zahid

Experiment 5

Lab Tasks

Experiment 5.py

Viva Questions

1. Regression is used when the output we want to predict is a continuous numerical value like price, temperature, or score, while classification is used when the output is a discrete category or class like spam or not spam, or what type of flower a plant is. In regression, the model predicts a number, whereas in classification, the model predicts a label. For example, predicting house prices is regression, while predicting if an email is spam is classification.

2. R^2 measures how well the model explains the variance in the data, with values closer to 1 indicating a better fit. It represents the proportion of the dependent variable that is predictable from the independent variables. RMSE measures the average size of the prediction errors in the same units as the target variable, with lower values indicating better model performance. R^2 tells us how much of the data the model can explain, while RMSE tells us how far off our predictions are on average.

3. K-Nearest Neighbors makes a prediction by finding the k closest data points in the training set to the new data point and then taking a vote among those neighbors. For

classification, it counts which class appears most among the k nearest neighbors and assigns that class to the new point. For regression, it takes the average of the target values of the k nearest neighbors. The distance between points is typically calculated using Euclidean distance, and the choice of k affects the model's performance.

4. We must test on data the model has not seen because we want to measure how well the model performs on new, unseen data in the real world. If we evaluate the model on the same data it was trained on, it might have simply memorized the examples instead of learning the underlying patterns, which gives us a false sense of accuracy. Testing on unseen data helps us detect overfitting and ensures that the model can generalize well to new situations beyond the training data.

5. Accuracy measures the proportion of correct predictions out of all predictions made, calculated as the number of correct predictions divided by the total number of predictions. However, accuracy can be misleading when dealing with imbalanced datasets where one class is much more frequent than the other. For example, if 95% of emails are not spam, a model that always predicts "not spam" would have 95% accuracy but would be completely useless for detecting spam. In such cases, precision, recall, and F1-score provide more meaningful evaluation.

6. Two algorithms that can do both regression and classification are Decision Trees and K-Nearest Neighbors (KNN). Decision Trees can be used for regression by averaging the target values in each leaf node, and for classification by taking the majority class in each leaf. KNN can be used for regression by averaging the target values of the k nearest neighbors, and for classification by taking the majority class among them. Both algorithms are versatile and can handle different types of prediction problems depending on how they are configured.

MCQ's

Q1. Predicting spam vs not-spam is:

(b) Classification

Q2. A perfect R^2 score is:

(c) 1

Q3. KNN predicts using the:

(b) k nearest points

Q4. Lower RMSE means:

(b) better model

Q5. Which is used for classification?

(b) Logistic Regression

Intern Name: Ghazi Muhammad Abdullah
Supervisor Name: Mr. Mohazin Zahid
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Domain: AI/ML Intern

Intern Signature: